SMARTLEARN — PROJECT UPDATE REPORT

C-Based Smart Learning & Quiz System

North South University | Project Progress Update | October 2026

1. Introduction

SmartLearn is a console-based educational application being developed in the C programming language. It aims to give students a simple environment for practicing questions and taking quizzes while storing learning and result information in text files. As a university team project, it provides practical experience with core programming concepts and basic file handling.

2. Work Completed So Far

The project has progressed from initial setup to a working C application structure with source code and text files for question data, student information, results, and topic statistics. The source code uses file operations such as fopen() to read and write records. Question records use a delimiter-based format to store question details, answer options, and the correct answer.

The application source includes student login/registration flows and question-based learning modes, including practice and quizzes. Recent code updates addressed file-reading behavior, including handling longer input lines, with the aim of making saved data easier to load reliably. Feature completeness and correctness still need to be confirmed through systematic testing.

3. Repository and Documentation Updates

The team maintains a shared public GitHub repository for source code and project documentation. The main C source file is located at src/main.c. The repository includes README.md, requirements.txt, TEST_CASES.md, PROJECT_PROGRESS.md, and project data/record files.

The project update presentation has been added to the others/ folder in PowerPoint format, with a PDF version intended to make slide previews easier in a browser. The repository structure will still be checked against the instructor’s required folders, including data/, support/, and others/.

4. Current Development Focus

The current priority is to compare the application with the intended requirements and verify the behavior of existing features. Testing should include registration and login, question loading, practice and quiz sessions, saving and reading records, and behavior when files are missing or contain invalid data. Test outcomes should be recorded accurately; a feature should not be described as fully working until its behavior has been checked.

5. Remaining Work and Next Steps

- Review the requirements and label each feature as complete, incomplete, or needing improvement.
- Run the test cases in TEST_CASES.md and update the file with actual results, including any failures and retests.
- Confirm that the repository follows the instructor’s required structure and that files are stored in the appropriate folders.
- Review README.md and requirements.txt so setup, compilation, execution, and any dependencies are clearly explained.
- Coordinate genuine team contributions so each member can explain the work they personally completed.
- Prepare the final report, final presentation, and short demonstration video after implementation and testing have been reviewed.

6. Conclusion

SmartLearn has established its C-programming foundation, file-based data handling, GitHub repository, and project update materials. The next stage is to validate the current behavior systematically, resolve gaps against the agreed requirements, and complete the documentation and final submission package. Final readiness should be confirmed only after required features have been tested and the team has checked the source code and supporting materials.

Verification note: This is a progress report. Confirm feature status against the latest source code and actual test results before submission.